

Chapter 24: Electromagnetic Waves

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PHY201
Lecture 24



Part I

Maxwell's Equations: Prediction of EM Waves

Outline of Part I: Maxwell's Equations: Prediction of EM Waves

A Historical Puzzle

Maxwell's Four Equations

Speed of Electromagnetic Waves

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Light as a Particle, as a Wave, and as a Puzzle

How can light travel through empty space?

Sound waves are described by oscillatory motion in **air**.

Water waves are described by oscillatory motion in **water**.

Experiments reveal that **light behaves as a wave**.

But light reaches us from the Sun through **empty space**; no matter is present.

So if **nothing** is there, **what** is in oscillatory motion?

Wave Types Compared

- ▶ **Sound:** oscillating pressure in matter.
- ▶ **Water waves:**
oscillating surface of a fluid.
- ▶ **Electromagnetic (EM) waves:**
oscillating *electric and magnetic fields*.



EM waves do *not* require matter.

The **fields** carry energy through space, not matter.

This was Maxwell's revolutionary insight.

Before Maxwell: Two Separate Sciences

The Historical Situation in the 1850s

- ▶ Electricity and magnetism were treated as completely separate subjects
- ▶ Coulomb, Ampère, and Faraday had each developed their own laws
- ▶ No one recognized electricity and magnetism as aspects of a single force
- ▶ Faraday had shown: changing $\vec{B}$ induces $\vec{E}$
- ▶ Ampère's law: electric currents create $\vec{B}$

The Asymmetry Maxwell Noticed

- ▶ Faraday: changing $B \Rightarrow E$ ✓
- ▶ Ampère: current $\Rightarrow B$ ✓
- ▶ Reverse of Faraday: changing $E \Rightarrow B$?
missing!

Maxwell argued that *symmetry demands* a changing electric field must also produce a magnetic field — completing the picture.

Outline of Part I: Maxwell's Equations: Prediction of EM Waves

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Speed of Electromagnetic Waves

Maxwell's Four Equations (Conceptual)

The Four Laws that Unify Electromagnetism

1. **Gauss's Law:** Electric charges produce electric fields.
2. **Gauss's Law for Magnetism:** No isolated magnetic monopoles have ever been observed — magnetic field lines always form closed loops.
3. **Faraday's Law:** A *changing* magnetic field induces an electric field.
4. **Ampère–Maxwell Law:** Magnetic fields are produced by (a) electric currents and (b) *changing* electric fields.

Maxwell's Critical Addition

Clause (b) of the Ampère–Maxwell Law was Maxwell's own contribution — the **displacement current** term. It gives the theory its self-sustaining wave solution.

Maxwell's Key Insight: The Missing Symmetry

Faraday + Maxwell = Mutual Feedback

$$\text{Faraday's Law: } \Delta \vec{B} / \Delta t \Rightarrow \vec{E}$$

$$\text{Ampère–Maxwell Law: } \Delta \vec{E} / \Delta t \Rightarrow \vec{B}$$

Why This Creates a Wave

- ▶ A changing $\vec{E}$ at one location creates $\vec{B}$ in nearby space
- ▶ That changing $\vec{B}$ creates $\vec{E}$ slightly farther away
- ▶ The disturbance propagates outward continuously
- ▶ **No medium is needed** — only fields

Crucial Distinction

Static fields do **not** regenerate one another.

Only *changing* (time-varying) fields drive the feedback loop.

What Is Actually Oscillating?

In an electromagnetic wave, the **electric and magnetic field values** oscillate at every point in space and time. No material substance moves along with the wave.

Sound Wave vs. EM Wave

- ▶ **Sound:** molecules oscillate back and forth in the direction of travel; *requires particles*
- ▶ **EM wave:** $\vec{E}$ and $\vec{B}$ values oscillate at each point in space; *requires fields only*

Removing all matter from space leaves the *fields* intact. EM waves travel through perfect vacuum because the fields themselves are the “medium.”

Important: EM waves are *transverse* — the field oscillations are *perpendicular* to the direction of travel.

Unlike sound, which is a longitudinal (compression) wave.

Outline of Part I: Maxwell's Equations: Prediction of EM Waves

A Historical Puzzle

Maxwell's Four Equations

Speed of Electromagnetic Waves

Speed of Electromagnetic Waves

Maxwell's Prediction — SI Unit: $\frac{\text{m}}{\text{s}}$

Solving Maxwell's equations produces a wave that travels at speed:

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

where $\mu_0 = 4\pi \times 10^{-7} \frac{\text{T}\cdot\text{m}}{\text{A}}$ and $\epsilon_0 = 8.85 \times 10^{-12} \frac{\text{C}^2}{\text{N}\cdot\text{m}^2}$.

Variable Definitions

- ▶ c : speed of EM waves in vacuum
- ▶ μ_0 : permeability of free space
- ▶ ϵ_0 : permittivity of free space

Historical Significance

Maxwell calculated $c \approx 3.00 \times 10^8 \frac{\text{m}}{\text{s}}$

The measured speed of light was *already known* to be $\approx 3.00 \times 10^8 \frac{\text{m}}{\text{s}}$.

Light is an electromagnetic wave.

Example: Speed of Light from Fundamental Constants

Predicting the Speed of Light

Using $\mu_0 = 4\pi \times 10^{-7} \frac{\text{T}\cdot\text{m}}{\text{A}}$ and $\epsilon_0 = 8.85 \times 10^{-12} \frac{\text{C}^2}{\text{N}\cdot\text{m}^2}$, calculate the predicted speed of electromagnetic waves.

$$\begin{aligned} c &= \frac{1}{\sqrt{\mu_0 \epsilon_0}} \\ &= \frac{1}{\sqrt{(4\pi \times 10^{-7})(8.85 \times 10^{-12})}} \\ &= \frac{1}{\sqrt{1.113 \times 10^{-17}}} \end{aligned}$$

$$c \approx 3.00 \times 10^8 \frac{\text{m}}{\text{s}}$$

This predicted value closely matched the measured speed of light — one of the most profound results in the history of physics.

Light, electricity, and magnetism are **one unified phenomenon**.

Check Your Understanding

What was Maxwell's crucial addition to Ampère's Law — the idea that was *not* already known from Faraday or Ampère?

Come to lecture to see the answer.

Check Your Understanding

Why was the equation

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \approx 3.00 \times 10^8 \frac{\text{m}}{\text{s}}$$

so important historically?

Come to lecture to see the answer.

Part II

Production and Geometry of Electromagnetic Waves

Outline of Part II: Production and Geometry of Electromagnetic Waves

Accelerating Charges and Radiation

EM Wave Geometry

Outline of Part II: Production and Geometry of Electromagnetic Waves

Accelerating Charges and Radiation

EM Wave Geometry

Which Charges Radiate?

Charge Motion and Radiation

- ▶ A **stationary** charge: creates a static $\vec{E}$, no $\vec{B}$, *no radiation*
- ▶ A charge at **constant velocity**: creates both $\vec{E}$ and $\vec{B}$, *no radiation*
- ▶ An **accelerating** charge: changing $\vec{E}$ and $\vec{B}$ *radiate energy outward*

Why Acceleration Is Required

- ▶ Only *changing* fields feed the mutual-regeneration loop
- ▶ Constant velocity: fields change shape but don't generate propagating waves
- ▶ Oscillating (accelerating) charges: create continuously changing fields $\Rightarrow$ sustained radiation

Common Misconception

A stationary charge does *not* produce radiation even though it has an electric field.

A charge moving at constant speed does *not* radiate.

Radiation requires *acceleration*.

Oscillating Dipole Antenna

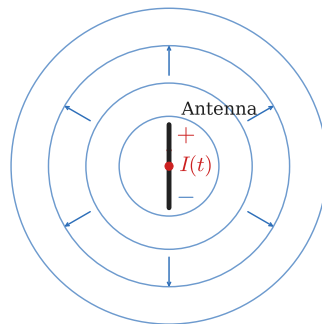
How a Radio Antenna Works

- ▶ AC current drives electrons up and down:
oscillating charge = accelerating charge
 $\Rightarrow$ radiation
- ▶ EM waves emitted at the driving frequency f
- ▶ Waves propagate outward carrying energy

Emitted wavelength: $\lambda = c/f$.

Half-wave antenna length: $\ell = \lambda/2$.

EM waves radiate outward



Outline of Part II: Production and Geometry of Electromagnetic Waves

Accelerating Charges and Radiation

EM Wave Geometry

Geometry of a Plane Electromagnetic Wave

Three Fundamental Geometric Facts

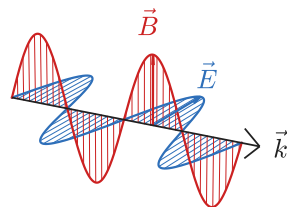
1. $\vec{E} \perp \vec{B}$ (the two fields are perpendicular to each other)
2. Both $\vec{E}$ and $\vec{B}$ are perpendicular to the direction of propagation
3. The fields oscillate *in phase* (peaks and zeros occur together)

Propagation Direction

The wave travels in the direction of:

$$\vec{E} \times \vec{B}$$

This cross-product points *away* from the source, carrying energy outward.



$$\vec{k} = \vec{E} \times \vec{B}$$

Electric and Magnetic Field Ratio

Field Amplitude Relationship — SI Unit: $\frac{\text{V}}{\text{m}} / \text{T}$

In a plane electromagnetic wave, the electric and magnetic field amplitudes satisfy:

$$c = \frac{E}{B}$$

where E and B are the instantaneous (or peak) field magnitudes.

Physical Interpretation

- ▶ E is numerically much larger than B (since $c \sim 3 \times 10^8 \frac{\text{m}}{\text{s}}$).
- ▶ Despite this, both fields carry *equal* amounts of energy.
- ▶ The ratio is fixed by the wave speed — it is not a free parameter.

From $c = E/B$:

$$B = \frac{E}{c}, \quad E = cB$$

Either field completely determines the other.

The EM Wave Equation

Frequency, Wavelength, and Wave Speed

All electromagnetic waves in vacuum obey:

$$c = f\lambda$$

where $c = 3.00 \times 10^8 \frac{\text{m}}{\text{s}}$ is the **same** for every EM wave in vacuum.

Key Insight

- ▶ All EM waves travel at *the same speed* in vacuum
- ▶ Frequency and wavelength adjust *inversely*
- ▶ High frequency $\Rightarrow$ short wavelength
- ▶ Low frequency $\Rightarrow$ long wavelength

💀 Common Misconception

Higher-frequency waves do **not** travel faster.

Radio waves and gamma rays both travel at c in vacuum.

Frequency determines energy per photon (quantum), not speed.

Example: Finding the Magnetic Field Amplitude

EM Wave Field Amplitudes

An EM wave has a peak electric field $E_0 = 1000 \frac{\text{V}}{\text{m}}$. Find (a) the peak magnetic field B_0 , and (b) the wavelength if the frequency is $f = 100 \text{ MHz}$.

$$\begin{aligned} \text{(a)} \quad B_0 &= \frac{E_0}{c} \\ &= \frac{1000}{3.00 \times 10^8} \end{aligned}$$

$$B_0 = 3.33 \times 10^{-6} \text{ T} = 3.33 \mu\text{T}$$

$$\begin{aligned} \text{(b)} \quad \lambda &= \frac{c}{f} \\ &= \frac{3.00 \times 10^8}{100 \times 10^6} \end{aligned}$$

$$\lambda = 3.00 \text{ m}$$

B_0 is tiny ($\sim \mu\text{T}$) yet both E and B fields carry equal energy.

$\lambda = 3 \text{ m}$ is in the FM radio band.

B_0 is small numerically because c is very large.

Check Your Understanding

An EM wave travels in the $+z$ direction. The electric field points in the $+x$ direction.

In which direction does the magnetic field point?

Come to lecture to see the answer.

Check Your Understanding

Why can electromagnetic waves travel through a perfect vacuum, when sound waves cannot?

Come to lecture to see the answer.

Check Your Understanding

Do radio waves (frequency ~ 1 MHz) travel slower than gamma rays (frequency $\sim 10^{20}$ Hz) in vacuum?

Come to lecture to see the answer.

Part III

The Electromagnetic Spectrum

Outline of Part III: The Electromagnetic Spectrum

The Spectrum as a Continuum

Radio Waves and Microwaves

Infrared, Visible, and Ultraviolet

X-Rays and Gamma Rays

Outline of Part III: The Electromagnetic Spectrum

The Spectrum as a Continuum

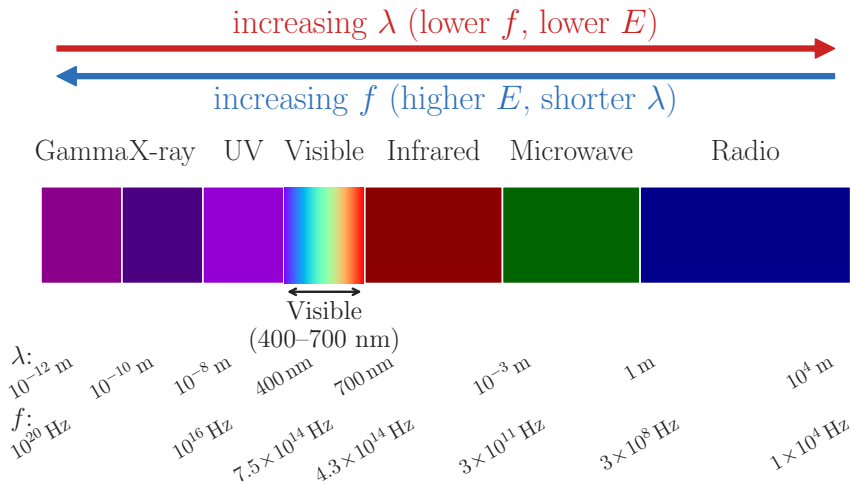
Radio Waves and Microwaves

Infrared, Visible, and Ultraviolet

X-Rays and Gamma Rays

The Electromagnetic Spectrum

All EM waves obey $c = f\lambda$ with the same c . They differ only in frequency (and therefore wavelength).



Three Unifying Trends Across the Spectrum

Frequency Determines Behavior

As frequency **increases** (wavelength decreases):

Energy per Photon

- ▶ Higher frequency $\Rightarrow$ higher photon energy
- ▶ Can ionize atoms and break chemical bonds
- ▶ Explains biological hazards of UV, X-rays, gamma rays

Spatial Resolution

- ▶ Shorter wavelength $\Rightarrow$ can resolve smaller structures
- ▶ X-rays image bone and soft tissue details
- ▶ Radio waves cannot resolve small objects

Information Bandwidth

- ▶ Higher frequency carriers $\Rightarrow$ more information per second
- ▶ Fiber optics (IR/visible) outperform radio
- ▶ 5G uses higher frequencies than 4G for this reason

Outline of Part III: The Electromagnetic Spectrum

The Spectrum as a Continuum

Radio Waves and Microwaves

Infrared, Visible, and Ultraviolet

X-Rays and Gamma Rays

Radio Waves

Properties

- ▶ Longest wavelengths in the EM spectrum
- ▶ Produced by oscillating currents in antennas
- ▶ Penetrate buildings and the lower atmosphere effectively
- ▶ Frequency range: $\sim 3 \text{ kHz}$ to 300 MHz

Applications

- ▶ AM/FM radio broadcasting
- ▶ Television transmission
- ▶ WiFi, Bluetooth, GPS
- ▶ MRI (uses radio-frequency pulses)
- ▶ Radar (overlaps with microwave)

Why radio for MRI?

MRI uses radio-frequency EM pulses to flip nuclear spins in hydrogen atoms. The body's own protons emit radio waves as they return to equilibrium.

No ionizing radiation is involved.

Microwaves

Properties

- ▶ Wavelengths from ~ 1 mm to ~ 30 cm
- ▶ Produced by magnetrons and klystrons
- ▶ Can penetrate clouds (useful for radar)

Applications

- ▶ **Microwave ovens:** 2.45 GHz excites water-molecule rotations, generating heat
- ▶ Radar: aircraft, weather, speed guns
- ▶ Satellite and cell phone communication
- ▶ Cosmic Microwave Background (CMB) astronomy

Microwave-Oven mechanism:

Water molecules are polar — they rotate to align with the rapidly oscillating $\vec{E}$ field. This rotational motion is heat.

Dry foods with little water heat slowly.

Outline of Part III: The Electromagnetic Spectrum

The Spectrum as a Continuum

Radio Waves and Microwaves

Infrared, Visible, and Ultraviolet

X-Rays and Gamma Rays

Infrared Radiation

Properties

- ▶ Wavelengths: $\sim 700 \text{ nm}$ to $\sim 1 \text{ mm}$
- ▶ Emitted by any object with temperature above 0 K
- ▶ Associated with molecular vibrations and thermal radiation
- ▶ Human body radiates primarily in the infrared

Applications

- ▶ Thermal imaging (medical, military)
- ▶ Remote controls; night vision
- ▶ Fiber optic communication (1550 nm)

The **greenhouse effect** involves IR.

Sunlight warms Earth's surface, which re-radiates as IR.

Greenhouse gases absorb this IR, trapping heat.

Visible Light

Properties

- ▶ Wavelengths: ~ 400 nm (violet) to ~ 700 nm (red)
- ▶ Only a narrow slice of the entire EM spectrum
- ▶ Produced by atomic electron transitions

Color Order: ROYGBIV

- ▶ **Red:** longest λ , lowest frequency
- ▶ **Violet:** shortest λ , highest frequency
- ▶ Higher frequency $\Rightarrow$ higher photon energy

Visible Light: Not Special

Visible light is *not* fundamentally different from radio waves or X-rays.

It is simply the narrow frequency range detectable by the human eye.

The same physical laws govern all EM radiation.

Ultraviolet Radiation

Properties

- ▶ Wavelengths: ~ 10 nm to ~ 400 nm
- ▶ Higher photon energy than visible light
- ▶ Can ionize molecules and break chemical bonds
- ▶ Classified: UV-A (400–315 nm), UV-B (315–280 nm), UV-C (<280 nm)

Beneficial Effects

- ▶ UV-B triggers vitamin D synthesis in skin
- ▶ UV-C kills bacteria (sterilization)
- ▶ Fluorescence and black-light applications

Hazards

- ▶ Damages DNA $\Rightarrow$ sunburn, cataracts, skin cancer
- ▶ UV-B and UV-C absorbed by the ozone layer
- ▶ Ozone depletion increases UV-B surface exposure

Ozone layer: absorbs most UV-B and virtually all UV-C, protecting life at Earth's surface.

Outline of Part III: The Electromagnetic Spectrum

The Spectrum as a Continuum

Radio Waves and Microwaves

Infrared, Visible, and Ultraviolet

X-Rays and Gamma Rays

X-Rays

Properties

- ▶ Wavelengths: ~ 0.01 nm to ~ 10 nm
- ▶ Produced by rapid deceleration of high-energy electrons (in an X-ray tube)
- ▶ Highly penetrating through soft tissue; absorbed by bone (high- Z nuclei)
- ▶ Ionizing radiation: can damage DNA

Applications

- ▶ Medical X-ray imaging and CT scans
- ▶ Dental X-rays
- ▶ X-ray crystallography (determining molecular structure)
- ▶ Airport security scanning

X-ray vs. gamma ray:

Distinguished by *origin*, not wavelength:

- ▶ X-rays: produced by **electrons** through *bremssstrahlung*:
German for “braking radiation”.
bremsen: to brake
der Strahl: ray (of light)
- ▶ Gamma rays: produced by **nuclear** processes

Overlapping wavelength ranges exist.

Gamma Rays

Properties

- ▶ $\lambda < 0.01 \text{ nm}$; produced by nuclear decay
- ▶ Most penetrating EM radiation
- ▶ Requires cm of lead or meters of concrete to shield
- ▶ Strongly ionizing — highly damaging to tissue

Applications

- ▶ Cancer radiation therapy
- ▶ PET scans; sterilization of equipment
- ▶ Astrophysics: gamma-ray bursts, neutron stars

Penetration Hierarchy

- ▶ **Radio:** no shielding needed
- ▶ **UV:** skin and clothing
- ▶ **X-rays:** lead aprons
- ▶ **Gamma:** cm of lead or m of concrete

A Bridge Toward Quantum Physics

Two Levels of Description for EM Waves

- ▶ **Classical:** intensity depends on amplitude squared (E_0^2)
- ▶ **Quantum:** energy is also carried in discrete packets — **photons**

Two Independent Quantities

- ▶ **Frequency f :** sets energy *per photon*
- ▶ **Amplitude E_0 :** sets total intensity
- ▶ Higher $f \Rightarrow$ each photon can ionize atoms

X-rays and radio waves can have the same intensity, yet X-rays are far more biologically dangerous because each X-ray photon carries enough energy to ionize an atom.

Key distinction:

Intensity = total energy per area per time

Photon energy = hf determines whether a single quantum can break a bond

Both matter for biological effects.

Check Your Understanding

Which color of visible light has the shortest wavelength?

Come to lecture to see the answer.

Check Your Understanding

Radiation with wavelength $\lambda = 0.1 \text{ nm}$ is used to image bone structure. What type of EM radiation is this?

Come to lecture to see the answer.

Check Your Understanding

Order these from **lowest** to **highest** frequency:

visible light infrared gamma rays microwaves
ultraviolet X-rays

Come to lecture to see the answer.

Check Your Understanding

If the ozone layer were significantly depleted, what would be the primary consequence at Earth's surface?

Come to lecture to see the answer.

Part IV

Energy in Electromagnetic Waves

Outline of Part IV: Energy in Electromagnetic Waves

Intensity and Field Amplitudes

Outline of Part IV: Energy in Electromagnetic Waves

Intensity and Field Amplitudes

Electromagnetic Waves Carry Energy

Energy Transport by EM Waves

- ▶ EM waves transfer energy from source to receiver without matter transport
- ▶ Energy is stored in the electric and magnetic fields themselves
- ▶ The **direction** of energy flow is $\vec{E} \times \vec{B}$ (same as propagation direction)
- ▶ Both fields contribute *equally* to the total energy density

Energy Density in Fields

$$u_E = \frac{1}{2}\epsilon_0 E^2 \quad (\text{electric})$$

$$u_B = \frac{B^2}{2\mu_0} \quad (\text{magnetic})$$

In an EM wave: $u_E = u_B$ at all times.

The equal partition of energy between $\vec{E}$ and $\vec{B}$ is a direct consequence of:

$$c = \frac{E}{B} = \frac{1}{\sqrt{\mu_0\epsilon_0}}$$

Intensity

Intensity I — SI Unit: $\frac{\text{W}}{\text{m}^2}$

Intensity is the average power delivered per unit area perpendicular to the direction of propagation:

$$I = \frac{P}{A}$$

Variable Definitions

- ▶ I : intensity ($\frac{\text{W}}{\text{m}^2}$)
- ▶ P : power (W)
- ▶ A : area through which wave passes (m^2)

Examples of Intensity

- ▶ Sunlight at Earth's surface:
 $\approx 1000 \frac{\text{W}}{\text{m}^2}$
- ▶ Laser pointer: $\sim 10^4$ – $10^6 \frac{\text{W}}{\text{m}^2}$
- ▶ Dim starlight: $\sim 10^{-10} \frac{\text{W}}{\text{m}^2}$

Average EM Wave Intensity

Intensity in Terms of Field Amplitudes — SI Unit: $\frac{\text{W}}{\text{m}^2}$

For a sinusoidal EM wave with peak fields E_0 and B_0 :

$$I_{\text{ave}} = \frac{c\epsilon_0 E_0^2}{2} = \frac{cB_0^2}{2\mu_0} = \frac{E_0 B_0}{2\mu_0}$$

Physical Interpretation

- ▶ Intensity scales as E_0^2 (amplitude squared)
- ▶ The factor $\frac{1}{2}$ comes from time-averaging $\sin^2(\omega t)$
- ▶ Doubling E_0 quadruples intensity

☠ Peak vs. Average

Since the instantaneous intensity oscillates between 0 and I_{max} :

$$I_{\text{max}} = 2 I_{\text{ave}}$$

Detectors typically measure I_{ave} .

Intensity Follows the Inverse-Square Law over Space

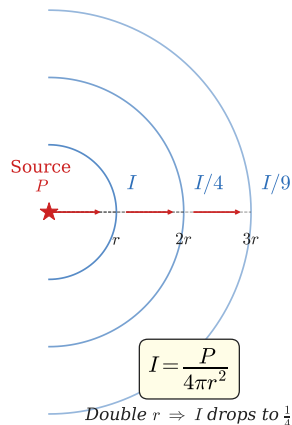
Point Source in Free Space

A source of total power P radiates uniformly in all directions. At distance r , the power is spread over a sphere of area $A = 4\pi r^2$:

$$I = \frac{P}{4\pi r^2}$$

Consequence

- ▶ Double the distance $\Rightarrow$ intensity falls to $\frac{1}{4}$
- ▶ Triple the distance $\Rightarrow$ intensity falls to $\frac{1}{9}$
- ▶ The *total* power P is conserved — it just spreads over a larger area



Example: Intensity and Field Amplitudes from a Lamp

Fields from a Lamp

100 W bulb, uniform radiation, $r = 1.00$ m: find I , E_0 , B_0 .

$$I = \frac{P}{4\pi r^2} = \frac{100}{4\pi(1)^2}$$

$$I \approx 7.96 \frac{\text{W}}{\text{m}^2}$$

$$E_0 = \sqrt{\frac{2I}{c\epsilon_0}}$$

$$= \sqrt{\frac{2(7.96)}{(3 \times 10^8)(8.85 \times 10^{-12})}}$$

$$E_0 \approx 77.5 \frac{\text{V}}{\text{m}}$$

$$B_0 = \frac{E_0}{c} \approx 2.58 \times 10^{-7} \text{ T}$$

$E_0 \approx 78 \frac{\text{V}}{\text{m}}$, $B_0 \approx 260 \text{ nT}$ at 1 m.

Intensity $\propto E_0^2$; fields fall as $1/r$.

At $r = 2$ m: intensity = $I/4$.

Example: Intensity from the Sun

Solar Radiation at Earth

Sun power $P = 3.85 \times 10^{26}$ W, Earth distance $r = 1.50 \times 10^{11}$ m: find I and E_0 .

$$I = \frac{P}{4\pi r^2}$$
$$= \frac{3.85 \times 10^{26}}{4\pi(1.50 \times 10^{11})^2}$$

$$I \approx 1360 \frac{\text{W}}{\text{m}^2}$$

$$E_0 = \sqrt{\frac{2I}{c\epsilon_0}}$$
$$= \sqrt{\frac{2(1360)}{(3.00 \times 10^8)(8.85 \times 10^{-12})}}$$

$$E_0 \approx 1012 \frac{\text{V}}{\text{m}}$$

Solar constant $\approx 1360 \frac{\text{W}}{\text{m}^2}$ — basis for solar panel design.

Atmosphere absorbs $\sim 30\%$, leaving $\sim 1000 \frac{\text{W}}{\text{m}^2}$ at Earth's surface.

Example: Using the Intensity Formula

Electric Field from Intensity

A microwave oven produces an average intensity of $I_{\text{ave}} = 5000 \frac{\text{W}}{\text{m}^2}$ inside the cooking cavity. Find the peak electric field E_0 and peak magnetic field B_0 .

$$E_0 = \sqrt{\frac{2 I_{\text{ave}}}{c \epsilon_0}} \quad \because I_{\text{ave}} = \frac{c \epsilon_0 E_0^2}{2}$$

$$= \sqrt{\frac{2(5000)}{(3.00 \times 10^8)(8.85 \times 10^{-12})}}$$

$$E_0 \approx 1940 \frac{\text{V}}{\text{m}}$$

$$B_0 = \frac{E_0}{c} = \frac{1940}{3.00 \times 10^8}$$

$$B_0 \approx 6.47 \mu\text{T}$$

Nearly $2000 \frac{\text{V}}{\text{m}}$ peak — that's why metal in a microwave sparks.

$$I_{\text{max}} = 2 I_{\text{ave}} = 10\,000 \frac{\text{W}}{\text{m}^2}.$$

Note: $E_0 \propto \sqrt{I}$, not I .

Check Your Understanding

A point source of light has power P . If you move from a distance r to a distance $3r$, by what factor does the intensity change?

Come to lecture to see the answer.

Check Your Understanding

The peak electric field E_0 of an EM wave is doubled. By what factor does the average intensity change?

Come to lecture to see the answer.

Check Your Understanding

At equal intensity, why are X-rays more biologically damaging than radio waves?

Come to lecture to see the answer.

Part V

Chapter Synthesis

Outline of Part V: Chapter Synthesis

The Unifying Theme

Outline of Part V: Chapter Synthesis

The Unifying Theme

The Unifying Theme of Chapter 24

Changing $\vec{E} \Leftrightarrow$ Changing $\vec{B} \Rightarrow$ self-propagating EM wave
 $\Rightarrow$ light, radio, X-rays, and all EM radiation are one phenomenon

The Four Pillars of Chapter 24

- ▶ **Prediction:** Maxwell predicted waves at speed c
- ▶ **Production:** accelerating charges radiate
- ▶ **Spectrum:** one continuum, organized by f
- ▶ **Energy:** $I \propto E_0^2$, falls as $1/r^2$

Historical Legacy

- ▶ 1865: Maxwell predicts EM waves
- ▶ 1888: Hertz detects radio waves experimentally
- ▶ Opens the modern era of wireless communication and quantum mechanics

Key Equations of Chapter 24

Wave Properties

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = f\lambda = \frac{E}{B}$$

Intensity

$$I = \frac{P}{4\pi r^2}$$

$$I_{\text{ave}} = \frac{c\epsilon_0 E_0^2}{2}$$

$$I_{\text{max}} = 2 I_{\text{ave}}$$

Spectrum (low $\rightarrow$ high f)

- ▶ Radio, Microwave, Infrared
- ▶ Visible: R $\rightarrow$ V = low $\rightarrow$ high f
- ▶ UV, X-ray, Gamma ray

Constants:

$$c = 3.00 \times 10^8 \frac{\text{m}}{\text{s}}$$

$$\mu_0 = 4\pi \times 10^{-7} \frac{\text{T}\cdot\text{m}}{\text{A}}$$

$$\epsilon_0 = 8.85 \times 10^{-12} \frac{\text{C}^2}{\text{N}\cdot\text{m}^2}$$

Looking Ahead

What Comes Next

- ▶ **Geometric Optics** (Ch. 25–26): reflection, refraction, lenses and mirrors
- ▶ **Wave Optics** (Ch. 27): interference and diffraction
- ▶ **Quantum Mechanics** (Ch. 29–31): EM waves carry energy in photons;
 $E_{\text{photon}} = hf$ (Note: this class only covers up to Ch. 28)

The Big Picture

Chapter 24 is the *bridge* between classical and modern physics.

Maxwell's equations are among the most fundamental and enduring laws in all of physics.

One phenomenon, many manifestations:

Radio, microwave, IR, visible, UV, X-rays, gamma rays — all are oscillating $\vec{E}$ and $\vec{B}$ fields, differing only in frequency.